Summary of the Base Paper and of the Network Estimators

Diebold and Yılmaz (2014) propose to measure how connected a set of assets is through the variance decomposition of an approximating VAR. The model is estimated and used to forecast each variable H periods ahead; d_{ij}^H is the share of the forecast-error variance of variable i due to shocks in variable j. Because shocks are usually correlated, the authors use the generalized variance decomposition of Koop et al. (1996) and Pesaran and Shin (1998), whose results do not depend on the ordering of the variables; its rows no longer sum to one, so they are normalized. The resulting matrix, augmented with its row and column sums, is the *connectedness table*. Each off-diagonal entry measures how much j affects i; the off-diagonal row sum gives how much a variable receives from the others ("from"), the column sum how much it transmits to them ("to"), and their difference whether it is a net transmitter or a net receiver. The average of all off-diagonal entries is the total connectedness of the system. Re-estimating the model over a rolling window shows how connectedness changes over time.

The main conceptual contribution of the paper is that the connectedness table is a weighted, directed network: "from" and "to" connectedness are the in- and out-degrees of each node, and total connectedness is the mean degree. The two directions also correspond to known systemic-risk measures: what a firm receives resembles exposure measures such as marginal expected shortfall, and what it transmits resembles contribution measures such as ΔCoVaR.

In the application, the authors study the daily realized volatilities of 13 large US financial institutions from May 1999 to April 2010, with a VAR(3), a 12-day horizon and a 100-day rolling window. Total connectedness is 78.3% in the full sample. Connectedness "from" others is tightly distributed (70–82%), whereas connectedness "to" others ranges from 53% to 106%, with Citigroup as the largest net transmitter. Over time the index shows two large cycles, around 2000–2003 and during the 2007–2009 crisis, when it reaches its maximum of 89.2% after the Lehman bankruptcy; the results are robust to the window, the horizon and the ordering of the variables. Our application changes the set of assets (emerging-market country funds), the object (returns instead of volatilities) and the frequency (weekly instead of daily).

Three limitations of the approach are relevant for our work. First, the VAR has N^2p coefficients and is often estimated with few observations. Second, every pair of variables ends up connected, even with a very small weight, so the network never has zero links. Third, the variance decomposition cannot tell whether assets move together because they affect each other or because they are all hit by the same shock. The two estimators below address these limitations.

NETS (Barigozzi and Brownlees, 2019) models the panel as a VAR in which both the autoregressive matrices and the precision matrix of the innovations (the inverse of their covariance matrix) are sparse, that is, most of their elements are zero. The dependence structure is then represented by networks. The Granger network is directed: there is a link from j to i when the coefficient on the lags of j in the equation for i is different from zero, so that j has predictive power for i. The contemporaneous network is undirected and is built from the partial correlations of the VAR innovations, which measure the relationship between two variables in the same period after controlling for all the others. The two combine into a long-run partial-correlation network. Estimation is carried out jointly by a lasso-type procedure that sets irrelevant links to zero, and, unlike the Diebold–Yilmaz approach, it requires no identifying assumptions on the shocks. In their application to the volatilities of 90 large US stocks, the authors find that the network captures part of the variability not explained by common volatility factors and improves out-of-sample forecasts. Its main limitation is that it does not model common factors explicitly: when a factor affects the whole panel, the dependence between series is no longer sparse and the central assumption of the model fails.

FNETS (Barigozzi et al., 2024) addresses this limitation with a factor-adjusted VAR. Each series is decomposed into a common component, driven by a small number of factors in a generalized dynamic factor model (Forni et al., 2000) and estimated by dynamic principal components, so that factors can affect each series with different lags; and an idiosyncratic component, modeled

as a sparse VAR. Since the idiosyncratic component is not observed, its parameters are estimated by an ℓ_1-regularized Yule–Walker estimator built from the idiosyncratic autocovariances implied by the estimated spectral density. The same three networks as in NETS are then estimated, and they reflect only the dependence that cannot be attributed to the common factors. The method also allows the common and idiosyncratic components to be forecast separately. The authors prove the consistency of the estimators as both the number of series and the sample size grow, also under heavier-than-Gaussian tails, and implement the procedure in the R package `fnets` (Owens et al., 2024). For our purpose, FNETS is the key method, because it separates the part of the connectedness among emerging markets that is due to global factors from the part that reflects links between the markets themselves.

1 Motivation

Emerging equity markets are exposed to two kinds of shocks. Some are global: changes in US monetary policy, in the dollar or in investors' appetite for risk move all of them at once, a pattern that Rey (2013) describes as a global financial cycle. Others are local, such as an election, a currency crisis or a change in economic policy in one country, and the question is whether, and through which countries, they spread to the rest. Telling the two apart matters for several reasons. For investors, the benefit of diversifying across emerging markets depends on how much of their co-movement is a common component that no portfolio of these markets can diversify away. For policy makers, a high degree of connectedness calls for different responses if it reflects bilateral transmission (contagion) or a common exposure to the same external shock. For the empirical literature, the distinction is an old one: Forbes and Rigobon (2002) showed that much apparent contagion during crises is high interdependence driven by common shocks and higher volatility.

The tools to measure connectedness have multiplied, and they do not all measure the same thing. The variance-decomposition approach of Diebold and Yılmaz (2014) summarizes how much of each market's forecast-error variance is due to shocks elsewhere; it is dense by construction and does not separate common from bilateral transmission. Sparse network estimators such as NETS (Barigozzi and Brownlees, 2019) keep only the links supported by the data, and FNETS (Barigozzi et al., 2024) first removes the common factor and then estimates the network of what is left. Applied to the same data, these methods can give quite different pictures of how connected markets are, who transmits and who receives, and how stable the network is. Before using a connectedness measure to draw conclusions about contagion, it is therefore worth knowing how much of the measured connectedness depends on the estimation method.

Our sample of 17 emerging markets offers clear examples of both kinds of shock. In the week of the Argentine primary elections of August 2019 the Argentine fund (ARGT) lost 28.4% (in log terms), its largest weekly fall in the sample: a large, country-specific shock. Seven months later, the COVID-19 crash of March 2020 produced the largest weekly losses of almost every market at the same time: 14 of the 17 markets recorded their worst week in March 2020. The Argentine episode illustrates the question we ask: whether a large local shock spreads to other markets, and whether the methods agree on that.

2 Introduction

This paper asks how measured connectedness among emerging equity markets changes with the method used to estimate it, and what share of it reflects a global common factor rather than idiosyncratic links between markets. We study weekly returns of US-listed country funds for 17 emerging markets in Latin America, Asia and EMEA, together with the S&P 500 as a global reference, from March 2011 to October 2026, and we estimate connectedness in four ways: the variance-decomposition measures of Diebold and Yılmaz (2014) on a VAR estimated by OLS;

the same measures on a sparse VAR estimated by the lasso with time-series cross-validation; NETS (Barigozzi and Brownlees, 2019), which estimates sparse Granger, contemporaneous and long-run partial-correlation networks; and FNETS (Barigozzi et al., 2024), which estimates the same networks after removing the common factor. All methods are estimated on the same rolling windows and mapped into comparable measures, and we compare them on three criteria: the density of the networks, the agreement between methods, and their stability over time.

We test three hypotheses. First, regularization alone should matter little for total connectedness at the weekly frequency, because the cross-market dependence of returns is mostly contemporaneous rather than lagged. Second, a substantial share of the connectedness measured by Diebold–Yilmaz should be due to a common factor, so that it falls markedly once the factor is removed, and its time variation should follow the weight of that factor. Third, sparse estimators should produce networks that are less dense, and also less stable from one window to the next, than the dense variance-decomposition network.

The results support the first two hypotheses and only partly the third. The lasso VAR sets almost all lag coefficients to zero yet gives nearly the same connectedness as Diebold–Yilmaz (80.0% against 81.0% on average over windows, correlation 0.98). NETS gives a lower level (61.5%) and keeps about half of the possible long-run links, while tracking the Diebold–Yilmaz series closely (correlation 0.89). FNETS finds a single common factor in every window, which explains on average 58% of the variance of the returns; removing it lowers connectedness by 41% on average, a share that is stable across window lengths, horizons, the exclusion of the S&P 500 and the exclusion of 2020. Over time, Diebold–Yilmaz connectedness and the weight of the common factor move together. The sparse networks are indeed much less dense, but stability depends on the measure: NETS connectedness jumps more from one window to the next than Diebold–Yilmaz, yet its strongest long-run links are more persistent; the idiosyncratic network of FNETS is both the sparsest and the least stable. South Africa, Korea and Taiwan are the main net transmitters and Turkey, Vietnam and Argentina the main net receivers in the dense and NETS systems; once the common factor is removed, Argentina is close to neutral.

Our contribution is to put the base paper's connectedness measures side by side with sparse and factor-adjusted network estimators on the same data, windows and metrics, for a set of emerging markets, and to quantify the part of measured connectedness that is a common factor. Section 3 describes the data, Section 4 the methods and comparison criteria, Section 5 the results, and Sections 6 and 7 discuss them and conclude. Robustness checks are in the appendix.

3 Data

Universe and sample. We use US-listed, USD-denominated country exchange-traded funds (ETFs) as tradable proxies for 17 emerging equity markets: Argentina (ARGT), Chile (ECH), Brazil (EWZ), Mexico (EWW) and Peru (EPU) in Latin America; South Korea (EWY), Taiwan (EWT), India (EPI), China (FXI), Thailand (THD), Indonesia (EIDO), Malaysia (EWM), Vietnam (VNM) and the Philippines (EPHE) in Asia; South Africa (EZA), Turkey (TUR) and Poland (EPOL) in EMEA. The S&P 500 ETF (SPY) is included as a global reference node. Funds were selected if their price history starts by 3 March 2011 (7-day tolerance), at most 5% of weeks are missing, and average daily traded volume is at least USD 3 million. Colombian funds (ICOL, GXG), INDA and MCHI did not pass the criteria; India and China are represented by EPI and FXI. The common sample runs from 11 March 2011 to 23 October 2026, i.e. 816 weeks, with no missing values.

Returns and frequency. Prices are dividend- and split-adjusted, so returns are total returns. We use weekly Friday-to-Friday log returns. Choosing the weekly frequency is a decision of ours, for three reasons: weekly returns retain some cross-market predictability, which is the dynamic dependence a VAR-based network is meant to capture; weekly aggregation mitigates the non-